Multi-Asset Quanto Pricing via C++ and Python Interface

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Quanto options

- A quanto option is written on an asset S_t, quoted in currency CCY1 with continuous dividend yield q, and pays in currency CCY2 at a fixed conversion rate. The holder receives the asset's performance in CCY2 without bearing exchange-rate risk.
- The foreign exchange (FX) rate X_t is the number of units of CCY1 per unit of CCY2.
- **Standard quanto**: A quanto call ($\phi = 1$) or put ($\phi = -1$) with strike K and expiry T pays $(\phi(S_T - K))^+$ units of CCY2.
- **Basket quanto**: With weights $w_j > 0$ on assets $S_{j,t}$, each quoted in its own currency, the basket quanto option pays $(\phi(\sum_j w_j S_{j,T} - K))^+$ units of CCY2.
- **Worst-of digital**: The worst-of digital call with fixing time T_1, expiry T_2 and strike ratio κ pays one unit of CCY2 when $\min_j S_{j,T_2}/S_{j,T_1} \geq \kappa$, and the put pays one unit otherwise.

Research question

Question

1. Price European quanto option from a closed-form formula, Monte Carlo simulation and a finite-difference partial differential equation (PDE).
2. How do the asset-FX and asset-asset correlations move single-asset, basket and worst-of digital quanto prices?

Finding

1. The three methods agree within their numerical error.
2. The asset-FX correlation raises quanto call prices and lowers put prices.
3. A quanto basket call is cheaper than the weighted average of its component calls.
4. Basket and worst-of digital call prices rise as the assets become more correlated.

Quanto drift

Under the CCY2 risk-neutral measure $\mathbb{Q}^d$,

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma_S\, dW_t^S, \qquad \frac{dX_t}{X_t} = \mu_X\, dt + \sigma_X\, dW_t^X, \qquad d\langle W^S, W^X\rangle_t = \rho\, dt,$$

with constant volatilities σ_S and σ_X and asset-FX correlation ρ.

- r_f and r_d are the continuously compounded interest rates of CCY1 and CCY2, and $Y_t = 1/X_t$ is the CCY2 price of one unit of CCY1.
- The discounted CCY2 values of a CCY1 deposit, $e^{(r_f - r_d)t} Y_t$ (Garman and Kohlhagen, 1983), and of the asset with dividends reinvested, $G_t = e^{(q-r_d)t} S_t Y_t$, are $\mathbb{Q}^d$ martingales.
- Setting the drift of G_t to zero gives the drift of S_t, the equivalent dividend yield and the quanto forward (Appendix B.1),

$$\mu = r_f - q + \rho\,\sigma_S\sigma_X, \qquad q' = r_d - r_f + q - \rho\,\sigma_S\sigma_X,$$
$$F^Q = \mathbb{E}^{\mathbb{Q}^d}[S_T] = S_0\, e^{(r_f - q + \rho\,\sigma_S\sigma_X)T}.$$

Closed-form price

$$V_\phi = \phi\, e^{-r_d T}\left[F^Q \Phi(\phi d_1) - K\Phi(\phi d_2)\right], \qquad d_{1,2} = \frac{\ln(F^Q/K) \pm \frac{1}{2}\sigma_S^2 T}{\sigma_S\sqrt{T}}$$

- V_ϕ is the price of the quanto call or put, and Φ the standard normal distribution function.
- This is the formula of Black and Scholes (1973) with interest rate r_d and dividend yield q'.

Correlation sensitivity

$$\frac{\partial V_\phi}{\partial \rho} = \phi\, \sigma_S \sigma_X\, T\, e^{-r_d T} F^Q \Phi(\phi d_1)$$

- The correlation enters V_ϕ only through F^Q, with $\partial F^Q / \partial \rho = \sigma_S \sigma_X\, T\, F^Q$.
- Hence the call price increases and the put price decreases in ρ.

- A positive ρ means the asset tends to rise when CCY1 weakens against CCY2.
- The CCY2 value $S_t Y_t$ must still grow at $r_d - q$, so the negative covariance between S_t and Y_t is offset by a higher drift of S_t itself.
- The quanto contract removes Y_t from the payoff and keeps that higher drift.

$$S_{t_i} = S_{t_{i-1}} \exp\left[\left(r_f(t_{i-1}, t_i) - q + \rho\,\sigma_S\sigma_X - \tfrac{1}{2}\sigma_S^2\right)\Delta t_i + \sigma_S\sqrt{\Delta t_i}\,\varepsilon_i\right], \qquad \varepsilon \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$$

- On fixing times $0 = t_0 < t_1 < \cdots < t_m$ the pricer advances $\ln S_t$ exactly.
- $r_f(t_{i-1}, t_i)$ is the CCY1 forward rate and $\Delta t_i = t_i - t_{i-1}$.

$$\frac{\partial V}{\partial t} + \left(\mu - \tfrac{1}{2}\sigma_S^2\right)\frac{\partial V}{\partial x} + \tfrac{1}{2}\sigma_S^2\frac{\partial^2 V}{\partial x^2} - r_d V = 0, \qquad V(T, x) = (\phi(e^x - K))^+$$

- By the Feynman-Kac theorem (Shreve, 2004, Chapter 6), the price $V(t, x)$ as a function of $x = \ln S$ solves this equation.
- The solver uses N time steps and a uniform grid of N spot nodes in x that spans 5 standard deviations around the forward and passes through $\ln S_0$.
- Each backward step applies a θ scheme to the central difference approximation of the drift and diffusion terms. We use $\theta = \frac{1}{2}$, the scheme of Crank and Nicolson (1947).

Multi-asset extension

$$\frac{dS_{j,t}}{S_{j,t}} = \mu_j\,dt + \sigma_j\,dW_t^j, \qquad \mu_j = r_j - q_j + \rho_j\,\sigma_j\sigma_{X_j}, \qquad d\langle W^j, W^k\rangle_t = \rho_{jk}\,dt,$$

where asset $j = 1, \dots, M$ is quoted in its own currency with interest rate r_j, dividend yield q_j and volatility σ_j, and its exchange rate X_j has volatility σ_{X_j} and correlation ρ_j with the asset.

- The drift of asset j involves only its own exchange rate, and a change of measure does not alter instantaneous correlations (Shreve, 2004, Chapter 5).
- The joint law of the asset prices under $\mathbb{Q}^d$ therefore depends on μ_j, σ_j and the correlation matrix $R = (\rho_{jk})$ alone. Correlations between exchange rates or between an asset and another asset's exchange rate do not enter.
- The basket quanto option and the worst-of digital are priced with it by Monte Carlo.

Implementation via C++ and Python interface

Python function	Method	Pricer	Payoff
`qFwdPrice`	Closed form	`quantoFwdPrice`	
`qEuroBS`	Closed form	`quantoEuropeanOptionBS`	
`qEuroBSMC`	Monte Carlo	`BsMcQuantoPricer`	`EuropeanCallPut`
`qEuroBSPDE`	PDE	`Pde1DSolver`	
`qBasketEuroBSMC`	Monte Carlo	`MultiAssetBsMcQuantoPricer`	`BasketEuropeanCallPut`
`qWorstofDigiBSMC`			`WorstOfDigitalCallPut`

Quanto functions.

- qflib is a C++20 library whose vectors and matrices are the Armadillo types.
- It is compiled into a static library and exposed to Python through an extension module written against the C interfaces of Python and NumPy.
- The script `quanto.py` produces every table and figure in the report by calling the C++ pricers from Python.

Inputs

Single asset		Basket and worst-of	
S_0, K	100	$S_{j,0}$, w_j	100, 1/3
T	one year	σ_j	20%, 25%, 30%
r_d, r_f	5%, 4%	q_j, σ_{X_j}, ρ_j	2%, 10%, 0.3
q	2%	ρ_{jk}	0.5
σ_S, σ_X	20%, 10%	K	100
ρ	0.3	κ, T_1, T_2	1, three months, one year
Monte Carlo	200,000 paths		
PDE	$N = 200$, $\theta = 0.5$, five standard deviations		

Inputs table.

- Both yield curves are assumed to be flat, with the US dollar as the payoff currency and the euro as the asset currency.
- PDE prices use $N = 200$ time steps and spot nodes.

Put-call parity

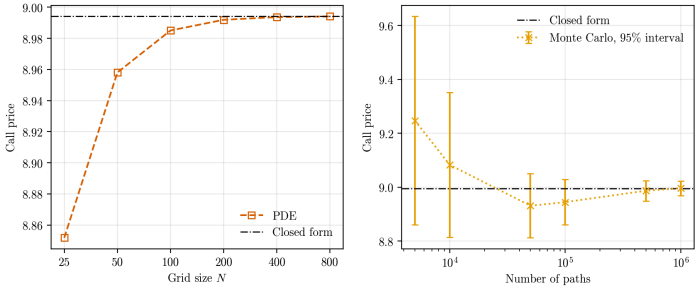
T	K	Call	Put	Residual
six months	80	21.03	0.24	1.3×10^{-12}
	100	6.19	4.92	1.6×10^{-12}
	120	0.86	19.09	1.9×10^{-12}
one year	80	22.41	0.88	1.9×10^{-12}
	100	8.99	6.49	1.9×10^{-12}
	120	2.61	19.13	2.0×10^{-12}
two years	80	24.97	2.05	1.4×10^{-11}
	100	13.03	8.20	1.4×10^{-11}
	120	6.02	19.28	1.4×10^{-11}

Put-call parity with PDE prices.
The residual is taken in absolute form.

$$V_+ - V_- = e^{-r_d T}\left(F^Q - K\right)$$

- V_+ and V_- are the call and put prices.
- PDE prices satisfy it over three expiries and three strikes.
- As $K \to 0$, $V_- \to 0$, so V_+ tends to the discounted quanto forward.

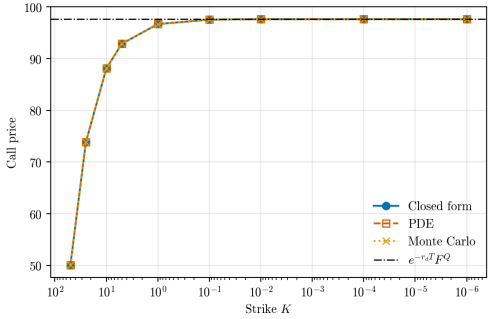
At-the-money call price

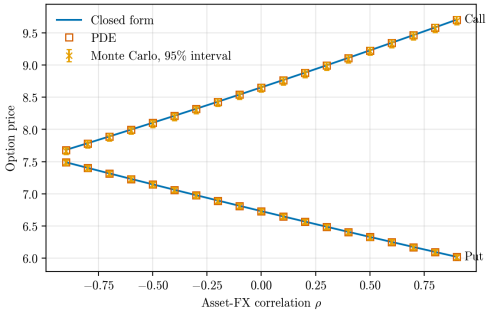


At-the-money call price against the PDE grid size and the number of Monte Carlo paths.

- The PDE error falls from 0.14 at $N = 25$ to 1.5×10^{-4} at $N = 800$ with a fitted slope of -1.99 on log scales, second order as expected for the Crank-Nicolson scheme.
- Every Monte Carlo estimate lies within 1.3 standard errors of the closed form, and at one million paths the error is 0.002 against a standard error of 0.014.

Single-asset quanto prices

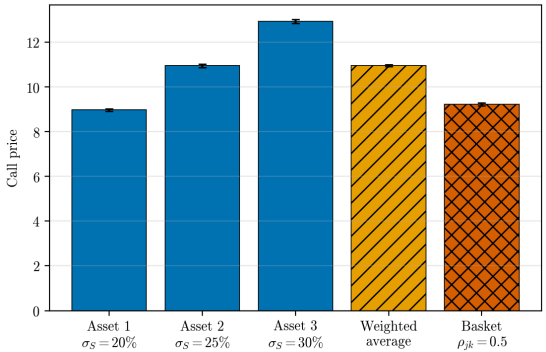




Single-asset quanto prices by the three methods, the call price as $K \to 0$ and the call and put prices against ρ.

- At $K = 10^{-6}$ the closed form and the PDE agree with the limit $e^{-r_d T} F^Q \approx 97.63$ to four decimals, and the Monte Carlo estimate of 97.62 lies 0.27 standard errors below it.
- For $\rho \in [-0.9, 0.9]$ the at-the-money call rises $7.68 \to 9.70$ and the put falls $7.49 \to 6.02$.

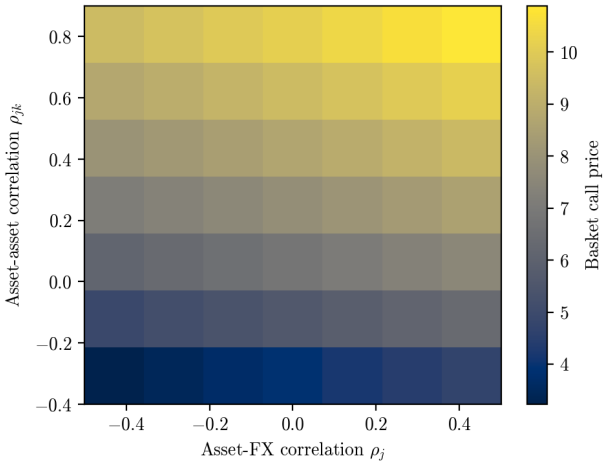
Quanto basket call prices



Component calls and the basket call. The error bars are 95% Monte Carlo intervals.

- At $\rho_{jk} = 0.5$ the basket volatility is 20.5% against an average component volatility of 25% (Appendix B.2).
- The basket call at 9.23 therefore lies much closer to the single 20% call at 8.98 than to the average of 10.96.

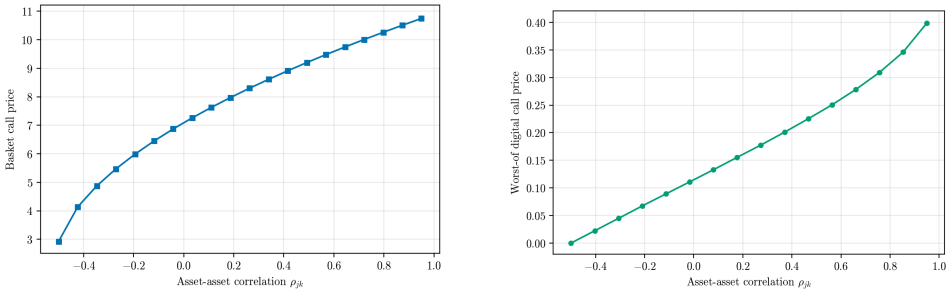
Basket call against both correlations



Quanto basket call prices.

- The common asset-FX correlation ρ_j runs from -0.5 to 0.5 and ρ_{jk} from -0.4 to 0.9.
- The basket call rises in both, from 3.23 to 10.88.

Prices against a common asset-asset correlation



Prices against a common asset-asset correlation, the basket call and the worst-of digital call.

- From $\rho_{jk} = -0.5$ the basket call rises from 2.93 to 10.75 and approaches the weighted average of 10.96.
- The worst-of digital call rises from zero to 0.40 (Appendix B.3), because the assets clear the strike together more often when they move together (Slepian, 1962).

Conclusion

- We price European quanto options by closed-form, Monte Carlo and PDE methods, and quanto basket calls and worst-of digitals by Monte Carlo, in C++ with a Python interface.
- The three methods agree within their numerical error and recover the discounted quanto forward at zero strike, and the PDE prices satisfy put-call parity.
- The asset-FX correlation enters through the drift, raising calls and lowering puts, and the multi-asset pricer is exact for baskets in several currencies.
- A quanto basket call is cheaper than the weighted average of its component calls because imperfect correlation lowers the basket volatility.
- Basket and worst-of digital calls rise as the assets become more correlated.

References

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Code is available at *GitHub*.

Appendix B.1. Quanto drift

One unit of CCY1 deposited at r_f is a CCY2 asset worth $e^{r_f t} Y_t$, so $e^{(r_f - r_d)t} Y_t$ is a $\mathbb{Q}^d$ martingale (Garman and Kohlhagen, 1983),

$$\frac{dY_t}{Y_t} = (r_d - r_f)\, dt - \sigma_X\, dW_t^X.$$

Holding one unit of the asset with dividends reinvested is worth $e^{qt} S_t Y_t$ in CCY2, so the discounted process $G_t = e^{(q - r_d)t} S_t Y_t$ is also a $\mathbb{Q}^d$ martingale,

$$\frac{dG_t}{G_t} = (q - r_d + \mu + r_d - r_f - \rho\, \sigma_S \sigma_X)\, dt + \sigma_S\, dW_t^S - \sigma_X\, dW_t^X.$$

Setting the drift to zero gives $\mu = r_f - q + \rho\, \sigma_S \sigma_X$.

Appendix B.2. Diversification

For $w_j \geq 0$ with $\sum_j w_j = 1$ and every realization, a call on a portfolio is worth no more than the portfolio of calls (Merton, 1973),

$$\Big(\sum_j w_j S_{j,T} - K \Big)^+ \leq \sum_j w_j \, (S_{j,T} - K)^+ .$$

The size of the gap comes from the dispersion of the basket. Matching the first two moments of B_T to a lognormal (Levy, 1992) gives a basket volatility σ_B with

$$\sigma_B^2 = \frac{1}{T} \ln \sum_{j,k} \omega_j \omega_k \, e^{\rho_{jk} \sigma_j \sigma_k T} \approx \sum_{j,k} \omega_j \omega_k \, \rho_{jk} \sigma_j \sigma_k \leq \Big(\sum_j \omega_j \sigma_j \Big)^2,$$

$$\omega_j = \frac{w_j F_j^Q}{\sum_k w_k F_k^Q}, \qquad F_j^Q = S_{j,0} \, e^{\mu_j T}.$$

For an at-the-money option, the closed form gives $V_+ \approx e^{-r_d T} F^Q \sigma_S \sqrt{T/2\pi}$, which increases and is nearly linear in volatility.

Appendix B.3. Asset-asset correlation

Since each $\ln(S_{j,T_2}/S_{j,T_1})$ is normal, the worst-of digital call price is a joint normal probability,

$$V^{\mathrm{WO}} = e^{-r_d T_2}\,\mathbb{P}\left(\xi_j \geq c_j \text{ for all } j\right), \qquad \xi \sim \mathcal{N}(0, R), \qquad c_j = \frac{\ln \kappa - (\mu_j - \frac{1}{2}\sigma_j^2)\delta}{\sigma_j \sqrt{\delta}},$$

where $\delta = T_2 - T_1$. This probability is nondecreasing in every ρ_{jk}, because the assets clear the strike together more often when they move together (Slepian, 1962). With $A_j = \{\xi_j \geq 0\}$ and the bivariate result of Sheppard (1899),

$$\mathbb{P}(A_j^c \cap A_k^c) = \frac{1}{4} + \frac{1}{2\pi} \arcsin \rho_{jk} \quad \Longrightarrow \quad \mathbb{P}\Big(\bigcap_j A_j\Big) = \frac{1}{8} + \frac{1}{4\pi} \sum_{j<k} \arcsin \rho_{jk}.$$

Hence, it vanishes at $\rho_{jk} = -\frac{1}{2}$, the smallest value for which R is positive semidefinite, and is convex for $\rho_{jk} > 0$, the shape of the worst-of digital call price.